

Week 4 Module:

Date: 09/07/2026

EASY LEVEL:

Problem Statement:

Given a positive integer **N**, determine the total number of digits present in the number.

Your task is to write a program that counts and prints the total number of digits.

Input Format

A single integer **N**.

Output Format

Print a single integer representing the total number of digits in **N**.

Constraints

$$1 \leq N \leq 10^{18}$$

Example 1

Input

12345

Output

5

Explanation

The number **12345** contains **5** digits.

Example 2

Input

7

Output

1

Explanation

The number **7** consists of only one digit.

Week 4 Module

Total Number of digit

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
long long n;
```

```
scanf("%lld", &n);
```

```
// Special case : 0 has 1 digit
```

```
if (n == 0)
```

```
{
```

```
printf("1\n");
```

```
return 0;
```

```
}
```

```
int count = 0;
```

```
while (n > 0)
```

```
{
```

```
n = n / 10;
```

```
// Integer division, remove last d
```

```
count ++;
```

```
// Increase digit count
```

```
}
```

```
printf("%d\n", count);
```

```
return 0;
```

```
}
```

Input

Example 1

12345

Output

5

Example 2

7 → Input

1 → Output

EXPLORER

OPEN EDITORS

GROUP 1

total number of digit.c

GROUP 2

C/C++ Compile Run output

WEEK 4 MODULE

.vscode

output

total number of digit.c

OUTLINE

TIMELINE

total number of digit.c

main()

```
1  #include <stdio.h>
2  int main()
3  {
4      long long n;
5      scanf("%lld", &n);
6      // Special case: 0 has 1 digit
7      if (n == 0) {
8          printf("1\n");
9          return 0;
10     }
11     int count = 0;
12     while (n > 0) {
13         n = n / 10;    // integer division, removes last digit
14         count++;      // increase digit count
15     }
16     printf("%d\n", count);
17     return 0;
18 }
```

C/C++ Compile Run output

```
cd "/Users/pulkitmittal230207icloud.com/week 4 module/output"
./"total number of digit"
• pulkitmittal230207icloud.com@192 week 4 module % cd "/Users/pulkitmittal230207icloud.com/week 4 module/output"
• pulkitmittal230207icloud.com@192 output % ./"total number of digit"
12345
5
• pulkitmittal230207icloud.com@192 output % cd "/Users/pulkitmittal230207icloud.com/week 4 module/output"
• pulkitmittal230207icloud.com@192 output % ./"total number of digit"
7
1
• pulkitmittal230207icloud.com@192 output %
```

0 0

week 4 module

Debug

Compile

Compile & Run

Ln 18, Col 2

Spaces: 4

UTF-8

LF

{ } C

macos-clang-arm64

Problem Statement

Given a positive integer **N**, reverse its digits and print the reversed number.

Your task is to write a program that reverses the given number.

Input Format

A single integer **N**.

Output Format

Print the reversed number.

Constraints

$$1 \leq N \leq 10^{18}$$

Example 1**Input**

12345

Output

54321

Explanation

Reversing the digits of **12345** gives **54321**.

Example 2**Input**

90010

Output

1009

Explanation

After reversing the digits, the leading zeros are removed automatically, resulting in **1009**.

reverse the number

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
long long n, rev = 0;
```

```
scanf("%lld", &n);
```

```
while (n > 0)
```

```
{
```

```
int digit = n % 10; // get last digit
```

```
rev = rev * 10 + digit; // build reversed number
```

```
n = n / 10; // removed last digit
```

```
}
```

```
printf("%lld\n", rev);
```

```
return 0;
```

```
}
```

Example 1

Input

12345

Output

54321

Example 2

Input

90010

Output

1009

EXPLORER

- > OPEN EDITORS
- > WEEK 4 MODULE
- > OUTLINE
- > TIMELINE

```
C reverse the number.c x
C reverse the number.c > main()
1  #include <stdio.h>
2  int main()
3  {
4      long long n, rev = 0;
5      scanf("%lld", &n);
6      while (n > 0)
7      {
8          int digit = n % 10;    // get last digit
9          rev = rev * 10 + digit; // build reversed number
10         n = n / 10;            // remove last digit
11     }
12     printf("%lld\n", rev);
13     return 0;
14 }
```

C/C++ Compile Run output x

```
cd "/Users/pulkitmittal230207icloud.com/week 4 module/output"
./"reverse the number"
• pulkitmittal230207icloud.com@192 week 4 module % cd "/Users/pulkitmittal230207icloud.com/week 4 module/output"
• pulkitmittal230207icloud.com@192 output % ./"reverse the number"
12345
54321
• pulkitmittal230207icloud.com@192 output % cd "/Users/pulkitmittal230207icloud.com/week 4 module/output"
• pulkitmittal230207icloud.com@192 output % ./"reverse the number"
90010
1009
○ pulkitmittal230207icloud.com@192 output %
```


Problem Statement

Given a positive integer **N**, calculate the sum of all its digits.

Your task is to write a program that computes and prints the sum of the digits.

Input Format

A single integer **N**.

Output Format

Print a single integer representing the sum of all digits.

Constraints

$$1 \leq N \leq 10^{18}$$

Example 1**Input**

12345

Output

15

Explanation

The sum of the digits is:

$$1 + 2 + 3 + 4 + 5 = 15$$

Example 2**Input**

9087

Output

24

Explanation

The sum of the digits is:

$$9 + 0 + 8 + 7 = 24$$

Sum of all its digit

```
# include <stdio.h>
```

```
int main()
```

```
{
```

```
long long n;
```

```
int sum = 0;
```

```
scanf("%lld", &n);
```

```
while (n > 0)
```

```
{
```

```
sum += n % 10; // add last digit
```

```
n = n / 10; // remove last digit
```

```
}
```

```
printf("%d\n", sum);
```

```
return 0;
```

```
}
```

Example 1

Input

12345

Output

15

Example 2

Input

9087

Output

24

week 4 module

EXPLORER

OPEN EDITORS

WEEK 4 MODULE

.vscode

output

reverse the number.c

sum of all its digits.c

total number of digit.c

OUTLINE

TIMELINE

sum of all its digits.c

```
1  #include <stdio.h>
2  int main()
3  {
4      long long n;
5      int sum = 0;
6      scanf("%lld", &n);
7      while (n > 0)
8      {
9          sum += n % 10; // add last digit
10         n = n / 10;    // remove last digit
11     }
12     printf("%d\n", sum);
13     return 0;
14 }
```

C/C++ Compile Run output

```
cd "/Users/pulkitmittal230207icloud.com/week 4 module/output"
./"sum of all its digits"
• pulkitmittal230207icloud.com@192 week 4 module % cd "/Users/pulkitmittal230207icloud.com/week 4 module/output"
• pulkitmittal230207icloud.com@192 output % ./"sum of all its digits"
12345
15
• pulkitmittal230207icloud.com@192 output % cd "/Users/pulkitmittal230207icloud.com/week 4 module/output"
• pulkitmittal230207icloud.com@192 output % ./"sum of all its digits"
9087
24
○ pulkitmittal230207icloud.com@192 output %
```

0 0

week 4 module

Debug

Problem Statement

Given a positive integer **N**, count how many **even** digits and how many **odd** digits are present in the number.

Your task is to write a program that prints the total count of even digits followed by the total count of odd digits.

Input Format

A single integer **N**.

Output Format

Print two integers separated by a space:

- Number of even digits
- Number of odd digits

Constraints

$$1 \leq N \leq 10^{18}$$

Example 1

Input

123456

Output

3 3

Explanation

Even digits are **2, 4, 6**.

Odd digits are **1, 3, 5**.

Therefore, the output is **3 3**.

Example 2

Input

908172

Output

3 3

Explanation

Even digits are **0, 8, 2**.

Odd digits are **9, 1, 7**.

Therefore, the output is **3 3**.

No of even and odd digit

#include <stdio.h>

int main()

{

long long n;

int even = 0, odd = 0;

scanf("%lld", &n);

while(n > 0)

{

int digit = n % 10;

if (digit % 2 == 0)

{

even++;

}

else

{ odd++;

}

n = n / 10;

}

printf("%d %d\n", even, odd);

return 0;

}

Example 1

Input

123456

Output

3 3

Example 2

Input

908172

Output

3 3

EXPLORER

OPEN EDITORS

WEEK 4 MODULE

- .vscode
- output
- no of even and odd digits.c
- reverse the number.c
- sum of all its digits.c
- total number of digit.c

OUTLINE

TIMELINE

```
C no of even and odd digits.c > ...
1  #include <stdio.h>
2  int main()
3  {
4      long long n;
5      int even = 0, odd = 0;
6      scanf("%lld", &n);
7      while (n > 0)
8      {
9          int digit = n % 10;
10         if (digit % 2 == 0)
11             even++;
12         else
13             odd++;
14         n = n / 10;
15     }
16     printf("%d %d\n", even, odd);
17     return 0;
18 }
```

C/C++ Compile Run output X

```
cd "/Users/pulkitmittal230207icloud.com/week 4 module/output"
./"no of even and odd digits"
• pulkitmittal230207icloud.com@192 week 4 module % cd "/Users/pulkitmittal230207icloud.com/week 4 module/output"
• pulkitmittal230207icloud.com@192 output % ./"no of even and odd digits"
123456
3 3
• pulkitmittal230207icloud.com@192 output % cd "/Users/pulkitmittal230207icloud.com/week 4 module/output"
• pulkitmittal230207icloud.com@192 output % ./"no of even and odd digits"
908172
3 3
○ pulkitmittal230207icloud.com@192 output %
```


Problem Statement

Given a positive integer **N**, find and print the largest digit present in the number.

Your task is to write a program that determines the maximum digit.

Input Format

A single integer **N**.

Output Format

Print the largest digit present in the given number.

Constraints

$$1 \leq N \leq 10^{18}$$

Example 1**Input**

472953

Output

9

Explanation

The digits are **4, 7, 2, 9, 5, 3**.

The largest digit among them is **9**.

Example 2**Input**

100248

Output

8

Explanation

The digits are **1, 0, 0, 2, 4, 8**.

The largest digit is **8**.

maximum digit

```
#include <stdio.h>
int main ()
{
    long long n;
    int maxDigit;
    scanf ("%lld", &n);
    maxDigit = n % 10;
    while (n > 0)
    {
        int digit = n % 10;
        if (digit > maxDigit)
        {
            maxDigit = digit;
        }
        n = n / 10;
    }
    printf ("%d\n", maxDigit);
    return 0;
}
```

Example 1

Input

472953

Output

9

Example 2

Input

100248

Output

8

EXPLORER

OPEN EDITORS

WEEK 4 MODULE

- .vscode
- output
- C maximum digit .c
- C no of even and odd digits.c
- C reverse the number.c
- C sum of all its digits.c
- C total number of digit.c

OUTLINE

TIMELINE

C maximum digit .c

```
C maximum digit .c > main()
1  #include <stdio.h>
2  int main()
3  {
4      long long n;
5      int maxDigit;
6      scanf("%lld", &n);
7      maxDigit = n % 10;
8      while (n > 0)
9      {
10         int digit = n % 10;
11         if (digit > maxDigit)
12         {
13             maxDigit = digit;
14         }
15         n = n / 10;
16     }
17     printf("%d\n", maxDigit);
18     return 0;
19 }
```

C/C++ Compile Run output

```
cd "/Users/pulkitmittal230207icloud.com/week 4 module/output"
./"maximum digit "
• pulkitmittal230207icloud.com@192 week 4 module % cd "/Users/pulkitmittal230207icloud.com/week 4 module/output"
• pulkitmittal230207icloud.com@192 output % ./"maximum digit "
472953
9
• pulkitmittal230207icloud.com@192 output % cd "/Users/pulkitmittal230207icloud.com/week 4 module/output"
• pulkitmittal230207icloud.com@192 output % ./"maximum digit "
100248
8
○ pulkitmittal230207icloud.com@192 output %
```


MEDIUM LEVEL:

Problem Statement

Given two positive integers **L** and **R**, count how many prime numbers are present in the inclusive range **[L, R]**.

A **prime number** is a natural number greater than **1** that has exactly two distinct positive divisors: **1** and itself.

Your task is to write a program that counts and prints the total number of prime numbers in the given range.

Input Format

Two integers **L** and **R** separated by a space.

Output Format

Print a single integer representing the total number of prime numbers in the given range.

Constraints

$$1 \leq L \leq R \leq 10^5$$

Example 1

Input

1 10

Output

4

Explanation

The prime numbers between **1** and **10** are: 2, 3, 5, 7

Hence, the total count is **4**.

Example 2

Input

11 20

Output

4

Explanation

The prime numbers between **11** and **20** are: 11, 13, 17, 19

Hence, the total count is **4**.

Total Number of prime number in left to right

```
#include <stdio.h>
```

```
int isPrime (int n)
```

```
{
```

```
    if (n < 2)
```

```
    {
```

```
        return 0;
```

```
    }
```

```
    for (int i = 2; i <= n; i++)
```

```
    {
```

```
        if (n % i == 0)
```

```
        {
```

```
            return 0;
```

```
        }
```

```
    }
```

```
    return 1;
```

```
}
```

```
int main()
```

```
{
```

```
    int L, R, count = 0;
```

```
    scanf("%d %d", &L, &R);
```

```
    for (int i = L; i <= R; i++)
```

```
    {
```

```
        if (isPrime(i))
```

```
        {
```

```
            count++;
```

```
        }
```

```
    }
```

```
    printf("%d\n", count);
```

```
    return 0;
```

```
}
```


Example 1

Input

1 10

Output

4

Example 2

Input

11 20

Output

4

EXPLORER

OPEN EDITORS

WEEK 4 MODULE

- .vscode
- output
- maximum digit.c
- no of even and odd digits.c
- reverse the number.c
- sum of all its digits.c
- total number of digit.c
- total number of prime numbersin left to right...

total number of prime numbersin left to right.c X

total number of prime numbersin left to right.c > isPrime(int)

```
1  #include <stdio.h>
2  int isPrime(int n)
3  {
4      if (n < 2)
5      {
6          return 0;
7      }
8      for (int i = 2; i * i <= n; i++)
9      {
10         if (n % i == 0)
11         {
12             return 0;
13         }
14     }
15     return 1;
16 }
17 int main()
18 {
19     int L, R, count = 0;
20     scanf("%d %d", &L, &R);
21     for (int i = L; i <= R; i++)
22     {
23         if (isPrime(i))
24         {
25             count++;
26         }
27     }
28     printf("%d\n", count);
29     return 0;
30 }
```

C/C++ Compile Run output X

```
cd "/Users/pulkitmittal230207icloud.com/week 4 module/output"
./"total number of prime numbersin left to right"
• pulkitmittal230207icloud.com@192 week 4 module % cd "/Users/pulkitmittal230207icloud.com/week 4 module/output"
• pulkitmittal230207icloud.com@192 output % ./"total number of prime numbersin left to right"
1 10
4
• pulkitmittal230207icloud.com@192 output % cd "/Users/pulkitmittal230207icloud.com/week 4 module/output"
• pulkitmittal230207icloud.com@192 output % ./"total number of prime numbersin left to right"
11 20
4
• pulkitmittal230207icloud.com@192 output %
```


Problem Statement

Given a positive integer **N**, find the sum of all its **proper divisors**.

A **proper divisor** of a number is a positive divisor of the number excluding the number itself.

Your task is to write a program that calculates and prints the sum of all proper divisors of **N**.

Input Format

A single integer **N**.

Output Format

Print a single integer representing the sum of all proper divisors of **N**.

Constraints

$$1 \leq N \leq 10^6$$

Example 1

Input

12

Output

16

Explanation

The proper divisors of **12** are:

1, 2, 3, 4, 6

Their sum is:

$$1 + 2 + 3 + 4 + 6 = 16$$

Example 2

Input

15

Output

9

Explanation

The proper divisors of **15** are:

1, 3, 5

Their sum is:

$$1 + 3 + 5 = 9$$

Sum of all proper divisions

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    long long n;
```

```
    long long sum = 0;
```

```
    scanf("%lld", &n);
```

```
    for (long long i = 1; i <= n/2; i++)
```

```
    {
```

```
        if (n % i == 0)
```

```
        {
```

```
            sum += 1;
```

```
        }
```

```
    }
```

```
    printf("%lld\n", sum);
```

```
    return 0;
```

```
}
```

Example 1

Input

12

Output

16

Example 2

Input

15

Output

9

EXPLORER

OPEN EDITORS

WEEK 4 MODULE

- .vscode
- output
- maximum digit .c
- no of even and odd digits.c
- reverse the number.c
- sum of all its digits.c
- sum of all proper divisors .c**
- total number of digit.c
- total number of prime numbers in left to right...

OUTLINE

TIMELINE

sum of all proper divisors .c > main()

```
1 #include <stdio.h>
2 int main()
3 {
4     long long n;
5     long long sum = 0;
6     scanf("%lld", &n);
7     for (long long i = 1; i <= n / 2; i++)
8     {
9         if (n % i == 0)
10        {
11            sum += i;
12        }
13    }
14    printf("%lld\n", sum);
15    return 0;
16 }
```

C/C++ Compile Run output X

```
cd "/Users/pulkitmittal230207icloud.com/week 4 module/output"
./"sum of all proper divisors "
pulkitmittal230207icloud.com@192 week 4 module % cd "/Users/pulkitmittal230207icloud.com/week 4 module/output"
pulkitmittal230207icloud.com@192 output % ./"sum of all proper divisors "
12
16
pulkitmittal230207icloud.com@192 output % cd "/Users/pulkitmittal230207icloud.com/week 4 module/output"
pulkitmittal230207icloud.com@192 output % ./"sum of all proper divisors "
15
9
pulkitmittal230207icloud.com@192 output %
```

Problem Statement

Given a positive integer **N**, print all the factors of the number in **ascending order**.

A factor of a number is an integer that divides the number completely without leaving any remainder.

Your task is to write a program that prints all the factors of **N**.

Input Format

A single integer **N**.

Output Format

Print all the factors of **N** in ascending order, separated by spaces.

Constraints

$$1 \leq N \leq 10^6$$

Example 1**Input**

12

Output

1 2 3 4 6 12

Explanation

The numbers **1, 2, 3, 4, 6**, and **12** divide **12** exactly without leaving any remainder.

Example 2**Input**

25

Output

1 5 25

Explanation

The numbers **1, 5**, and **25** are the only factors of **25**.

print all factor in ascending order

```
#include <stdio.h>
int main()
{
    int n;
    scanf("%d", &n);
    for(int i = 1; i <= n; i++)
    {
        if(n % i == 0)
        {
            printf("%d ", i);
        }
    }
    return 0;
}
```

Example 1

Input

12

Output

1 2 3 4 6 12

Example 2

Input

25

Output

1 5 25

EXPLORER

OPEN EDITORS

WEEK 4 MODULE

- .vscode
- output
- maximum digit .c
- no of even and odd digits.c
- print all factors in ascending order.c
- reverse the number.c
- sum of all its digits.c
- sum of all proper divisors .c
- total number of digit.c
- total number of prime numbersin left to rig...

OUTLINE

TIMELINE

print all factors in ascending order.c

```
C print all factors in ascending order.c > main()
1  #include <stdio.h>
2  int main()
3  {
4      int n;
5      scanf("%d", &n);
6      for (int i = 1; i <= n; i++)
7      {
8          if (n % i == 0)
9          {
10             printf("%d ", i);
11         }
12     }
13     return 0;
14 }
```

C/C++ Compile Run output

```
cd "/Users/pulkitmittal230207icloud.com/week 4 module/output"
./"print all factors in ascending order"
pulkitmittal230207icloud.com@192 week 4 module % cd "/Users/pulkitmittal230207icloud.com/week 4 module/output"
pulkitmittal230207icloud.com@192 output % ./"print all factors in ascending order"
12
1 2 3 4 6 12
pulkitmittal230207icloud.com@192 output % cd "/Users/pulkitmittal230207icloud.com/week 4 module/output"
pulkitmittal230207icloud.com@192 output % ./"print all factors in ascending order"
25
1 5 25
pulkitmittal230207icloud.com@192 output %
```